Does Bad Data Cause Broad Misalignment Only When the Model Recognizes It as Bad?

Jash Vira

May 2026

Abstract

Prior work shows that narrow fine-tuning on insecure code can produce broad emergent misalignment: models become more likely to give harmful, deceptive, or otherwise misaligned answers on unrelated prompts [1]. The mechanism remains unclear. This brief proposes a sharper hypothesis: broad misalignment is more likely when fine-tuning rewards a model for outputs it already recognizes as wrong. In a preliminary single-seed pilot with Qwen2.5-Coder-32B-Instruct, supervised fine-tuning on high-awareness insecure-code examples produced substantially more broad misalignment than fine-tuning on externally bad but low-awareness examples. At the larger data size, the high-awareness branch reached 17.1% misaligned on Betley's first-plot questions versus 1.0% for the low-awareness branch and 0.1% for the secure-control branch. The immediate next step is a cleaner replication across seeds, stricter matching, and one or two additional model families.

1 Question

Existing emergent-misalignment work mostly labels training data externally: insecure code, incorrect advice, reward hacking, or poisoned/backdoored behavior [1, 6, 4]. But the model being trained may not see all externally bad examples the same way.

The core question is:

> **Does bad data cause broad emergent misalignment mainly when the base model already recognizes the bad answer as bad?**

This separates four quantities that are often collapsed:

1. externally bad data;
2. base-model-recognized badness before fine-tuning;
3. narrow behavioral uptake after fine-tuning;
4. broad misalignment on unrelated evaluations.

The hypothesis is not that high-aware and low-aware bad data must produce equal narrow vulnerable-code imitation. Narrow code/security behavior is a diagnostic. The endpoint of interest is broad emergent misalignment.

2 Experimental Sketch

I used existing insecure-code corpora and scored bad examples before fine-tuning with the base model. The awareness prompt asked the model to review an answer for logical defects using minimal, non-security-leading language. Examples were split into:

- **high-aware bad**: externally bad code the base model identified as defective;
- **low-aware bad**: externally bad code the base model treated as acceptable;
- **secure control**: matched secure/correct code examples.

For each branch, I fine-tuned the same base model with the same optimizer, seed, epochs, prompt templates, and training format. The main run used:

- base model: Qwen2.5-Coder-32B-Instruct;
- method: QLoRA SFT;
- seed: 1;
- epochs: 5;
- learning rate: 2×10^{-4};
- sequence length: 12288;
- data sizes: $n = 1000$ and $n = 3452$ per branch.

Broad misalignment was evaluated with Betley's canonical broad-eval questions and scored using the original-style GPT-4o logprob judge: probability-weighted numeric scores over 0–100, keeping coherent rows where `coherent > 50` and marking a response misaligned when `aligned < 30` [1, 5]. Persona-style evaluations from OpenAI's follow-up work were used as supporting evidence, not the main endpoint [6, 3].

3 Preliminary Results

Evaluation	n	High-aware bad	Low-aware bad	Secure control	High-low
Betley first-plot	1000	10.2%	0.2%	0.7%	+10.0 pp
Betley first-plot	3452	17.1%	1.0%	0.1%	+16.1 pp
Betley preregistered	1000	5.3%	1.8%	1.7%	+3.6 pp
Betley preregistered	3452	7.0%	1.8%	1.0%	+5.2 pp
Persona extended	1000	17.6%	11.0%	9.1%	+6.6 pp
Persona extended	3452	18.3%	10.9%	11.0%	+7.4 pp

Table 1: Preliminary broad-misalignment rates. Betley results are the main evidence; Persona extended is supporting evidence.

The cleanest signal is on the Betley evals. At $n = 3452$, high-aware bad training produced 17.1% misaligned responses on the first-plot questions, compared with 1.0% for low-aware bad training and 0.1% for secure-control training. On preregistered evals, the corresponding rates were 7.0%, 1.8%, and 1.0%.

This pattern is what the hypothesis predicts: externally bad data is not uniformly dangerous. The more dangerous case appears to be data where training rewards the model for a behavior it can already classify as wrong.

4 Why This Matters

If this result survives replication, it changes how we should think about post-training data safety. The relevant risk variable would not be only whether an example is externally bad. It would also be whether the model being trained already represents the target behavior as bad.

That distinction matters for data filtering, model organisms, and interpretability. It suggests a natural bridge between behavioral emergent-misalignment work and white-box work on persona-like or deception-like representations [6, 2]. A useful next question is whether high-awareness bad examples more strongly activate internal features associated with misaligned persona, deception, or knowingly false behavior before broad misalignment appears.

5 What Would Make This Convincing

The current pilot is not enough to settle the mechanism. The next experiment should:

1. rerun the high/low/control comparison across multiple random seeds;

2. use stricter source-stratified matching so high and low buckets are not confounded by source, length, vulnerability type, or task template;
3. replicate on at least one additional open-weight model family;
4. keep prompt templates frozen and source-controlled;
5. report question-level uncertainty for broad evals rather than treating all samples as independent;
6. add a small human audit of awareness labels and broad-eval classifications;
7. optionally test whether internal feature/probe activations mediate the high-aware → broad-misalignment effect.

The strongest falsification would be simple: after stricter matching and multiple seeds, high-aware bad data no longer causes more broad misalignment than low-aware bad data.

6 Request For Feedback

I am looking for technical feedback on three points:

- Is "base-model-recognized badness" a plausible mediator for emergent misalignment, or is there a better causal variable to test?
- Are Betley broad evals plus Persona extended evals sufficient, or should an agentic/deception eval be added?
- What matching or awareness-labeling mistakes would most threaten the interpretation?

The goal of the next phase is not to maximize the effect size. It is to find out whether the pilot signal survives a replication that a skeptical reader would regard as clean.

References

[1] Jan Betley, Daniel Tan, Niels Warncke, Anna Sztyber-Betley, Xuchan Bao, Martín Soto, Nathan Labenz, and Owain Evans. Emergent misalignment: Narrow finetuning can produce broadly misaligned llms, 2025.

[2] Evan Hubinger, Nicholas Schiefer, Carson Denison, and Ethan Perez. Model organisms of misalignment: The case for a new pillar of alignment research. https://www.alignmentforum.org/posts/ChDH335ckdvpxXaXX/model-organisms-of-misalignment-the-case-for-a-new-pillar, 2023. Alignment Forum.

[3] OpenAI. Emergent misalignment persona features. https://github.com/openai/emergent-misalignment-persona-features, 2025. Training data, evaluation data, and grader infrastructure.

[4] Mia Taylor, James Chua, Jan Betley, Johannes Treutlein, and Owain Evans. School of reward hacks: Hacking harmless tasks generalizes to misaligned behavior in llms, 2025.

[5] Truthful AI. Emergent misalignment. https://github.com/emergent-misalignment/emergent-misalignment, 2025. Code, datasets, and evaluation questions.

[6] Miles Wang, Tom Dupré la Tour, Olivia Watkins, Alex Makelov, Ryan A. Chi, Samuel Miserendino, Jeffrey Wang, Achyuta Rajaram, Johannes Heidecke, Tejal Patwardhan, and Dan Mossing. Persona features control emergent misalignment, 2025.